



Highly Tough and Thermally Stable Polylactide Blends Compatibilized with Glycidyl Methacrylate-Grafted Polypropylene

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To attain eco-friendly and sustainable polylactide (PLA) materials possessing highly enhanced toughness, thermal stability, and processability without significant loss in elastic modulus, for the first time, PLA-dominant blends with 1–30 wt% glycidyl methacrylate-grafted polypropylene (PPGMA) loadings are fabricated via an efficient masterbatch melt-compounding process. For the purpose, PPGMA is fabricated via in situ grafting reaction of PP with GMA and styrene. The scanning electron microscope images reveal that PLA/PPGMA blends do not show recognizable phase-separated domains, unlike immiscible PLA/PP blends. The Fourier-transform infrared spectroscopic and melt-rheological analyses support the presence of specific interactions between PLA and PPGMA as well as the compatibilizing effect of PPGMA-g-PLA formed during the melt-compounding. The thermal analyses demonstrate that PPGMA component accelerates the crystallization of PLA in the blends and that the thermal decomposition temperatures of PLA/PPGMA blends are higher than those of neat PLA and PPGMA components. The dynamic mechanical analysis shows that a maximum storage modulus is attained for PLA-dominant blend with 30 wt% PPGMA. Noticeably, the impact strength ($\approx 305.6 \text{ J m}^{-1}$) of PLA-dominant blend with only 5 wt% PPGMA loading is almost three times higher than that ($\approx 111.6 \text{ J m}^{-1}$) of neat PLA and it is very comparable to the value ($\approx 316.9 \text{ J m}^{-1}$) of neat PP.

have gained great attention from industry and academia to replace common petroleum-based plastics including polyethylene (PE), polypropylene (PP), polystyrene (PS), poly(vinyl chloride), and poly(ethylene terephthalate) (PET).^[1–3] Poly(lactic acid) or polylactide (PLA), an aliphatic polyester derived from renewable resources, is one of the most promising biodegradable and thermoplastic polymeric materials because of its good combined characteristics of mass production, compostability, and mechanical properties of Young's modulus and stiffness.^[4–7] PLA and its decomposition products, H_2O and CO_2 , have great appeal in the field of environmental protection because they are nontoxic or noncarcinogenic to the human body. In addition, PLA significantly reduces its environmental footprint, which can be achieved relatively easily through improvement in farming resources, efficiency increase in various production steps, and usage of renewable energy in the conversion process. The life cycle assessments that quantify the environmental sustainability reported that the nonrenewable energy use

1. Introduction

Plastics, semiorganic materials that come from oil or petroleum, are indispensable in our daily life because they are widely utilized in all sectors of life including textiles, electronics, healthcare products, toys, food packaging/containers, automotive parts, and many other household and industrial goods. Nonetheless, in recent decades, with growing environmental problems and diminishing petroleum resources, environmentally benign and renewable source-based polymers

(NREU) and global warming potential (GWP) of PLA pellets are $30.55\text{--}54.1 \text{ MJ kg}^{-1}$ and $0.5\text{--}2.02 \text{ kg CO}_2 \text{ eq kg}^{-1}$, respectively, which are far lower than petroleum-based polymers, such as PET (82.3 MJ kg^{-1} , $3.49 \text{ kg CO}_2 \text{ eq kg}^{-1}$), PS (87.4 MJ kg^{-1} , $3.40 \text{ kg CO}_2 \text{ eq kg}^{-1}$), high-density polyethylene (76.7 MJ kg^{-1} , $1.9 \text{ kg CO}_2 \text{ eq kg}^{-1}$), and PP (73.4 MJ kg^{-1} , $2.0 \text{ kg CO}_2 \text{ eq kg}^{-1}$).^[8] However, PLA presents several limitations in practical applications that require high heat resistance and durability, because of intrinsic brittleness, low thermal strain temperatures, high prices, and slow crystallization rates.^[9,10]

To overcome the shortcomings of PLA, various organic and inorganic fillers have been introduced as reinforcing materials to develop PLA-dominant blends and composites.^[5,11–23] The stereocomplexation ability of PLA isomers has been also proposed as an effective method for enhancing the thermal and mechanical properties.^[24,25] Some researchers have explored the potential of blending PLA with conventional polymers to improve performance and lower prices by investigating the blends in terms of miscibility, morphology, thermal, and mechanical properties. In particular, extensive research has been conducted

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to toughen PLA through melt-blending, copolymerization, and plasticization.^[26,27] Zhu and co-workers reported the preparation and characterization of fully biodegradable PLA/starch granule blends toughened with epoxidized soybean oil (ESO) as a bio-based reactive compatibilizer.^[28] They found that the elongation at break and impact strength of the ESO-compatible PLA/starch granule blends could be effectively improved by using maleic anhydride (MA)-grafted starch granules owing to the ready reactions between epoxy groups of ESO, MA groups of starch granules, and carboxylic end groups of PLA. Favis and Zolali have reported the toughening of cocontinuous PLA/linear low density PE (LLDPE) blends in which poly(ϵ -caprolactone), ethylene-methyl acrylate copolymer (EMA), or ethylene-methyl acrylate-glycidyl methacrylate copolymer (EMA-GMA) is percolated at the interface of the PLA/LLDPE blends as the intermediate rubbery phase.^[26]

Isotactic polypropylene (PP), a commercial petroleum-based polymer, is widely used in commodity goods, due to its low density, cost-effectiveness, chemical resistance, easy processability, good mechanical toughness, and satisfactory heat resistance.^[29] However, PP does not decompose itself. To solve these environmental problems, it can be easily manufactured by blending with a biodegradable polymer. Blending PLA with PP could achieve a semibiocomposite with biodegradable property, reinforced mechanical properties, and acceptable performance-cost combination.^[30] Since the melting temperature of PP is comparable to that of PLA and even higher than those of PE and elastomers, PP is considered as an ideal component for attaining toughened PLA blends manufactured by a facile and efficient melt-compounding. Deri and co-workers reported the rheological and mechanical properties by blending PLA with PP using a single screw extruder.^[31] They found that the tensile stresses and strains at break of the blends are far lower than the values expected from the mixing rule owing to the incompatibility between PLA and PP in the blends, although Young's moduli of the blends are proportional to the PLA content. To solve the immiscibility issue of PLA/PP blends, a compatibilizer can be incorporated. The compatibilizer reduces the interfacial energy between the two polymers, consequently aids in dispersion and increases compatibility.^[32] Among compatibilizing agents, a block or graft copolymer compatible with the blend component may be used. Alternatively, there is a method of increasing the compatibility of the interface by incorporating a reactive functional group into the polymer. Kim and co-workers reported the influences of compatibilizers, i.e., maleic anhydride-grafted PP (PP-g-MAH) and maleic anhydride-grafted styrene-ethylene/butylene-styrene copolymer (SEBS-g-MAH), and hydrolysis on the tensile and impact strength, interfacial tension, and morphology of PP/PLA (80/20 by wt%) blends.^[33] They found that a maximum tensile strength was attained for the PP/PLA blends with 3 phr PP-g-MAH before hydrolysis and the impact strength of the blends before and after hydrolysis increased with the SEBS-g-MAH content. It was thus concluded that the SEBS-g-MAH is an effective impact modifier to enhance the impact strength of the PP/PLA blends.

Glycidyl methacrylate (GMA) with an epoxy-functional group is widely used as a reactive compatibilizing agent by grafting to olefin-based polymers such as PE and PP, which increases adhesion properties with excellent reactivity and low polarity.^[34–36]

It is thus conjectured that the epoxy groups of GMA-grafted PP (PPGMA) can react or interact with the carboxyl/hydroxyl end groups of PLA chains, resulting in enhancing the toughness of PLA without significant loss of stiffness and thermal properties. In this study, for the first time, PLA-dominant blends with relatively small amounts of PPGMA (1–30 wt%) were fabricated by an efficient reactive melt-compounding technique, in which PPGMA-g-PLA can be formed as an in situ compatibilizing and/or interactive component. The morphology, rheological, thermal, and mechanical properties of PLA/PPGMA blends were investigated in terms of the PPGMA content as well as the reactivity and intermolecular interactions between PPGMA and PLA.

2. Experimental Section

2.1. Materials

A commercial PLA (2002D, NatureWorks LLC) with M_w of $\approx 112\,600\text{ g mol}^{-1}$, polydispersity index of 1.46, D-isomer content of 4.2 mol%, and glass transition temperature (T_g) of 58.5 °C was used as the dominant blend component. Isotactic PP (M_w of $\approx 670\,000\text{ g mol}^{-1}$ and M_n of $\approx 250\,000\text{ g mol}^{-1}$), glycidyl methacrylate (GMA, 97% purity), styrene (>99% purity), and benzoyl peroxide (BPO) as a free radical initiator were purchased from Sigma-Aldrich Com.

2.2. Preparation of PPGMA

PPGMA was prepared by in situ melt-compounding of neat PP with GMA, styrene, and BPO initiator in a co-rotating intermeshing twin-screw extruder (BA11, Bautek Com.) at a screw speed 50 rpm and a temperature profile from the feeding zone of 175 °C to the die zone of 195 °C. Before melt-compounding, 50 g PP powder was physically pre-mixed with 10 wt% GMA, 10 wt% styrene, and 0.2 wt% BPO. Styrene was used as a comonomer to inhibit the decomposition of PP chains and to improve the grafting efficiency of GMA. Then, the mixture was fed into the extruder at a temperature range of 175–195 °C. The in situ melt-compounded product was purified by dissolving in xylene at 120 °C and precipitating into acetone at room temperature to remove unreacted components, such as GMA and styrene. The PPGMA precipitate was filtered, washed with acetone, and dried under vacuum at 80 °C. The overall schematic procedure to fabricate PPGMA by the melt-compounding is shown in Figure 1.

2.3. Preparation of PLA/PPGMA Blends

Before fabricating PLA/PPGMA blends through masterbatch-based melt-compounding, neat PLA and PPGMA were dried at 50 °C overnight under vacuum. First, a masterbatch of PLA/PPGMA (70/30 by wt%) was prepared by melt-compounding using a co-rotating intermeshing twin-screw extruder (BA11, Bautek Com.) operated at a screw speed of 50 rpm. The temperature profile of the extruder barrels was controlled to be 175, 175, 185, 185, 195, and 195 °C from the feeding zone to the die. Second, PLA/PPGMA blends with 1–30 wt% PPGMA

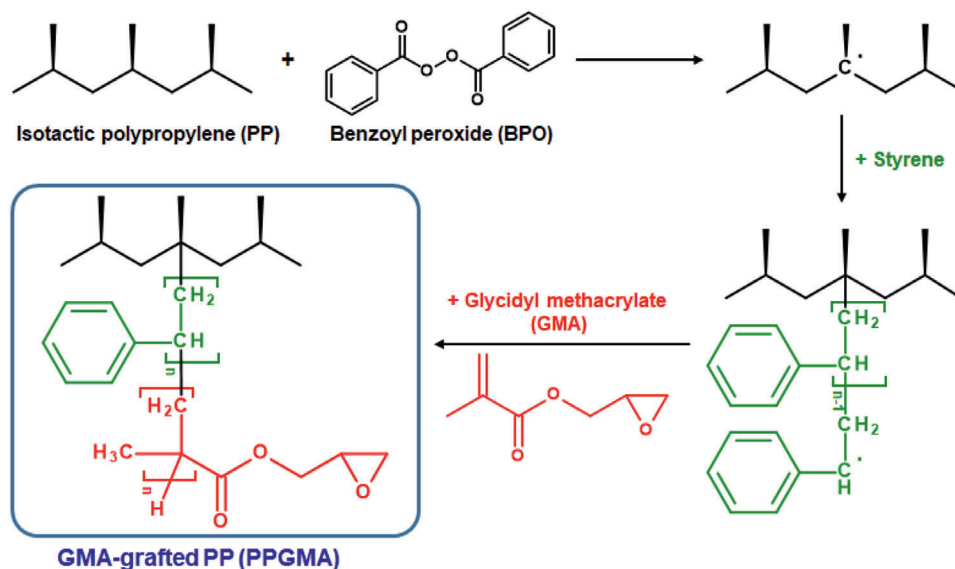


Figure 1. Schematic procedure to fabricate PPGMA by reactive melt-compounding.

contents were prepared by melt-compounding of the masterbatch chip with pristine PLA chips, chopping with a pelletizer, and drying overnight at 50 °C under vacuum. The final blends were named as PLA/PPGMA x , where x indicates the content of PPGMA by wt%. For comparison, a PLA/PP blend (70/30 by wt%) was prepared by the same melt-compounding process and it was named as PLA/PP30. For structural and property characterization, PLA/PPGMA and PLA/PP blends with thicknesses of 0.5 mm and 1.0 mm were prepared by compression-molding at 180 °C for 5 min, melt-quenching into iced-water, and drying in a vacuum oven at room temperature for several hours.

2.4. Characterization of PLA/PPGMA Blends

The morphologies of fractured surfaces of PLA/PPGMA and PLA/PP blends were characterized using a focused ion beam scanning electron microscope (FIB-SEM, LYRA3 XMU) operated at an accelerating voltage of 5 kV. For the SEM experiments, the blend samples were cryo-fractured in liquid nitrogen and the fractured surfaces were coated with osmium for 30 s in a vacuum chamber.

The molecular interactions between PLA and PPGMA in the blends were characterized by using a Fourier-transform infrared spectroscopy (FT-IR) spectrometer (Nicolet iS10, Thermo Fisher Scientific). FT-IR spectra were obtained in the wavenumber range of 400–4000 cm^{-1} .

The rheological properties of PLA/PPGMA blends in a melt-state were analyzed by using a rheometer (MCR 102, Anton Paar Com.) with a parallel plate geometry with 25.0 mm diameter and 1.0 mm gap. Dynamic frequency sweep measurements were conducted at 240 °C in the angular frequency range of 0.1–100 rad s^{-1} and constant shear strain of 2% to ensure the characterization that was in the linear viscoelastic region. Time sweep measurements were conducted at different temperatures of 180, 220, and 240 °C for 15 min at a constant shear rate of 0.05 s^{-1} .

The thermal transition properties of PLA blends with different PPGMA contents were investigated by using a differential scanning calorimeter (DSC 6000, PerkinElmer Inc.). For DSC experiments, each blend sample of 9–10 mg was sealed in an aluminum pan. The DSC heating and cooling thermograms were obtained in the temperature range of 30–200 °C at a scanning rate of 10 °C min^{-1} under nitrogen atmosphere.

The thermal stability of PLA/PPGMA blends and their components was characterized with aid of a thermogravimetric analyzer (TGA 4000, PerkinElmer Inc.), which was operated in the temperature range of 30–800 °C at a heating rate of 10 °C min^{-1} under a nitrogen gas flow.

The dynamic mechanical properties of the blends were performed using a dynamic mechanical analyzer (DMA 8000, PerkinElmer Inc.) in a tensile mode. All samples with 20 mm $\times$ 0.5 mm $\times$ 0.5 mm were tested by applying a constant strain of 0.1% in the temperature range from –50 to 150 °C at a heating rate of 3 °C min^{-1} .

The impact strength of PLA/PPGMA blends and their components with dimensions of 80.0 mm $\times$ 15.0 mm $\times$ 3.0 mm was tested by using a digital Izod impact tester (QMESYS, QM750A).

3. Results and Discussion

3.1. Morphological Characterization

The SEM images of the fractured surfaces of PLA, PP, PLA/PP30, and PLA/PPGMA blends are represented in **Figure 2**. The neat PLA showed a smooth surface due to its brittle and glassy fracture characteristic (Figure 2A), while the neat PP exhibited a relatively rough feature owing to its semicrystalline tough failure feature (Figure 2B). For the PLA/PP30 blend, a distinct phase-separated morphology with dispersed PP domains was observed owing to the immiscibility between PLA and PP (Figure 2H). In cases of PLA/PPGMA blends, ductile and rough fractured surfaces with only tiny domains and/or over the blend matrices were detected,

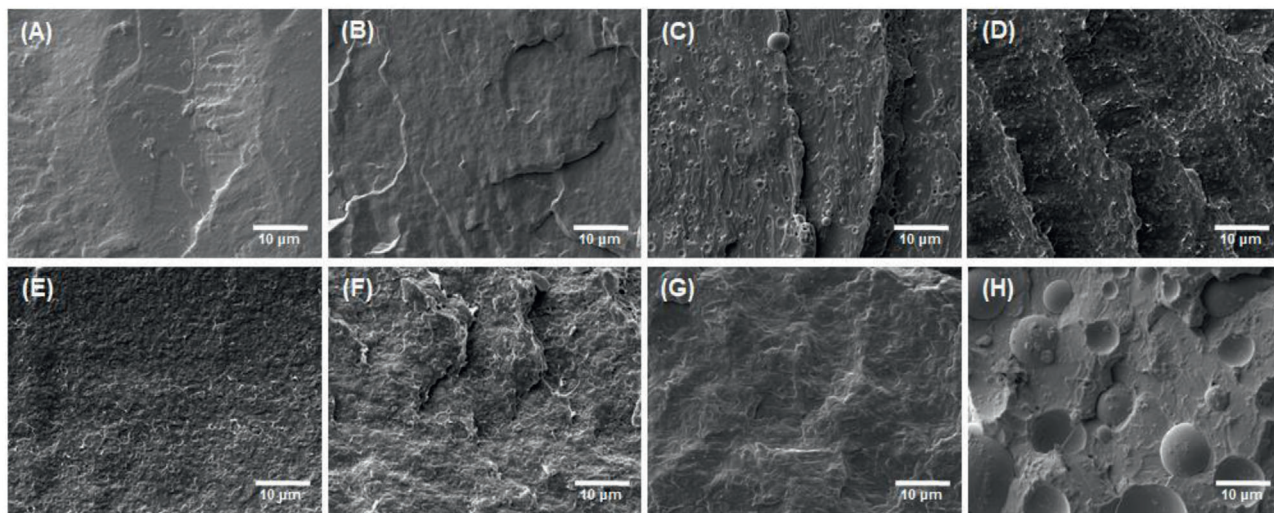


Figure 2. SEM images of fractured surfaces of A) PLA, B) PP, C) PLA/PPGMA1, D) PLA/PPGMA5, E) PLA/PPGMA10, F) PLA/PPGMA20, G) PLA/PPGMA30, H) PLA/PP30.

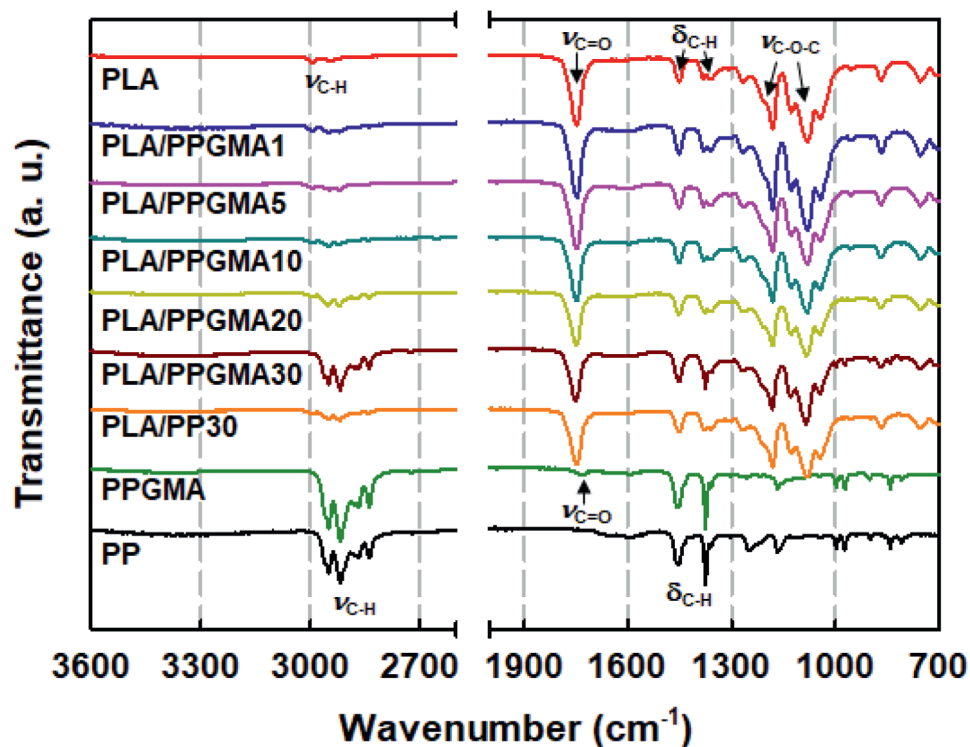


Figure 3. FT-IR spectra of PLA, PPGMA, PP, and their blends.

as shown in Figure 2C–G, which is caused by the compatibilizing effect of PPGMA at the interface of immiscible PLA and PP segments. These results indicate that the epoxy functional groups of PPGMA can react or interact with carboxyl and hydroxyl groups of PLA in the PLA/PPGMA blends, which results in the formation of copolymers between PLA and PPGMA during the in situ melt-compounding and reduces the interfacial tension between PLA and PP components.

3.2. Molecular Interactions of PLA/PPGMA Blends

To identify the specific interactions between PLA and PPGMA in the blends at a molecular level, FT-IR spectra of PLA/PPGMA blends and their components (PLA, PP, and PPGMA) are shown in Figure 3 and the assignment of typical vibrational bands is summarized in Table 1. For the neat PLA, characteristic bands of C–H stretching ($\nu_{\text{C-H}}$), C=O stretching ($\nu_{\text{C=O}}$), asymmetric

Table 1. Characteristic vibrational bands of PLA, PPGMA, PP, and their blends.

Sample code	Vibrational bands [cm ⁻¹]			
	$\nu_{\text{C=O}}$	$\delta_{\text{C-H}}$	$\nu_{\text{C-O-C}}$	$\nu_{\text{C-O-C}}$
PLA	1741	1444	1178	1072
PLA/PPGMA1	1740	1440	1176	1072
PLA/PPGMA5	1741	1444	1178	1074
PLA/PPGMA10	1743	1446	1178	1076
PLA/PPGMA20	1745	1446	1178	1078
PLA/PPGMA30	1745	1446	1178	1078
PLA/PP30	1741	1444	1178	1074
PPGMA	1734	1440	–	–
PP	–	1442	–	–

C–H bending ($\delta_{\text{C-H,asym}}$), symmetric C–H bending ($\delta_{\text{C-H,sym}}$), and C–O–C stretching ($\nu_{\text{C-O-C}}$) vibrations were detected at 2995–2946, ≈ 1741 , ≈ 1444 , and 1178–1072 cm⁻¹, respectively. In case of the neat PP, characteristic vibrational bands associated with C–H stretching ($\nu_{\text{C-H}}$), asymmetric C–H bending ($\delta_{\text{C-H,asym}}$), and symmetric C–H bending ($\delta_{\text{C-H,sym}}$) were observed at 2925–2815, 1442, and 1373 cm⁻¹, respectively.^[37] The overall FT-IR spectrum of PPGMA was similar with that of the neat PP, except for the appearance of new C=O stretching ($\nu_{\text{C=O}}$) band at ≈ 1734 cm⁻¹. The result supports that the GMA was successfully grafted on PP chains during the reactive melting-compounding. In cases of the PLA/PPGMA blends, the characteristic bands of both PLA and PPGMA were detected and their apparent intensities were dependent on the relative compositions of the blends. In addition, the band position of C=O stretching ($\nu_{\text{C=O}}$) of the PLA/PPGMA blends was found to shift from 1740 to 1745 cm⁻¹ with increasing the PPGMA content from 1 to 30 wt%. It demonstrates the

presence of grafting reactions and/or specific interactions between epoxy/ester groups of PPGMA and hydroxy/carboxyl/ester groups of PLA in the blends, as represented schematically in Figure 4.

3.3. Rheological Properties

The rheological characterization of polymer blends is known as an important analytical method to identify the processability of materials. For immiscible polymer blends, their melt-flow behavior is complex due to the strain sensitivity of blend morphology as well as the presence of several phases with different flow characteristics.^[38] To compare the rheological behavior of the compatibilized PLA/PPGMA30 with the immiscible PLA/PP30 blend, the time-dependent melt viscosity changes were measured at a constant shear rate of 0.05 s⁻¹ and different temperatures of 180–240 °C, as shown in Figure 5. As expected, the time-dependent melt viscosity values of both blends decreased with increasing the temperature. On the other hand, at a given temperature, the melt viscosities of PLA/PPGMA30 blend were far higher than those of PLA/PP30 blend. Furthermore, the difference in the melt viscosities between PLA/PPGMA30 and PLA/PP30 blends was higher from 55% to 114%, as the temperature increased from 180 to 240 °C. It is also noteworthy that the melt viscosity of PLA/PPGMA30 blend at 240 °C increased slightly with the time, whereas the viscosity values of PLA/PP30 blend kept decreasing with the time, irrespective of the temperatures of 180–240 °C. It is believed that the time-dependent viscosity changes of PLA/PP and PLA/PPGMA blends are associated with the morphology change as well as reactive compatibilization. In the case of the immiscible PLA/PP30 blend, phase separation proceeds with time in the molten state, and such phase separation is promoted at higher temperatures. This is in good agreement with the experimental results that the higher the temperature, the greater the decrease in viscosity over time.

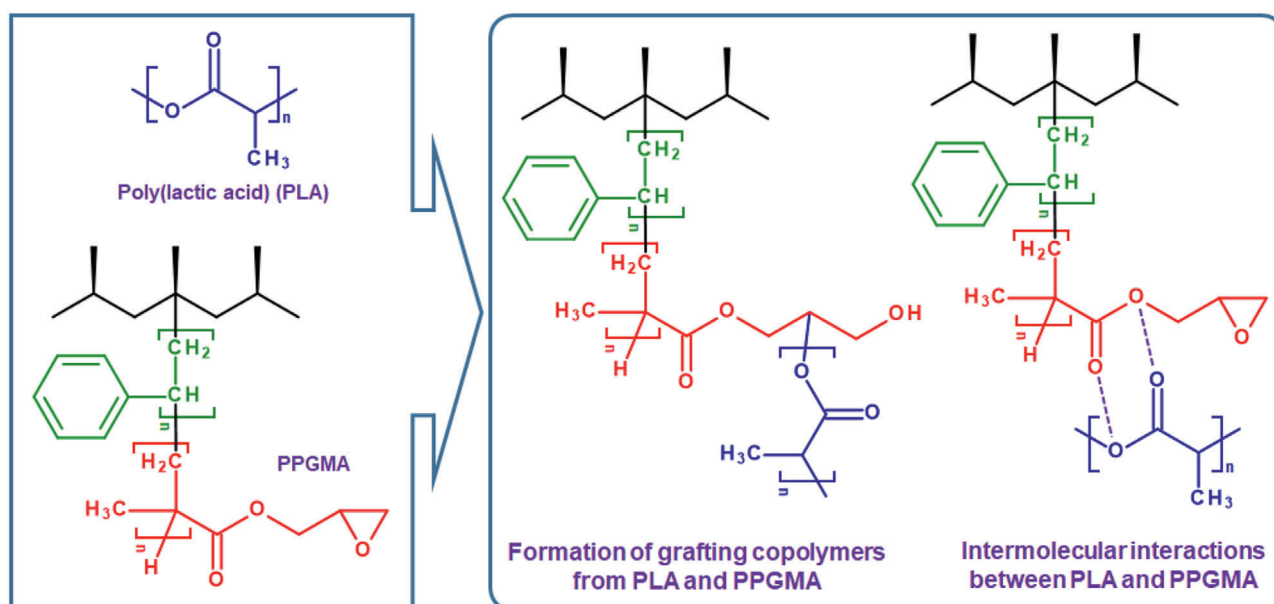


Figure 4. Scheme for the grafting copolymer formation and intermolecular interactions between PLA and PPGMA during reactive melt-compounding.

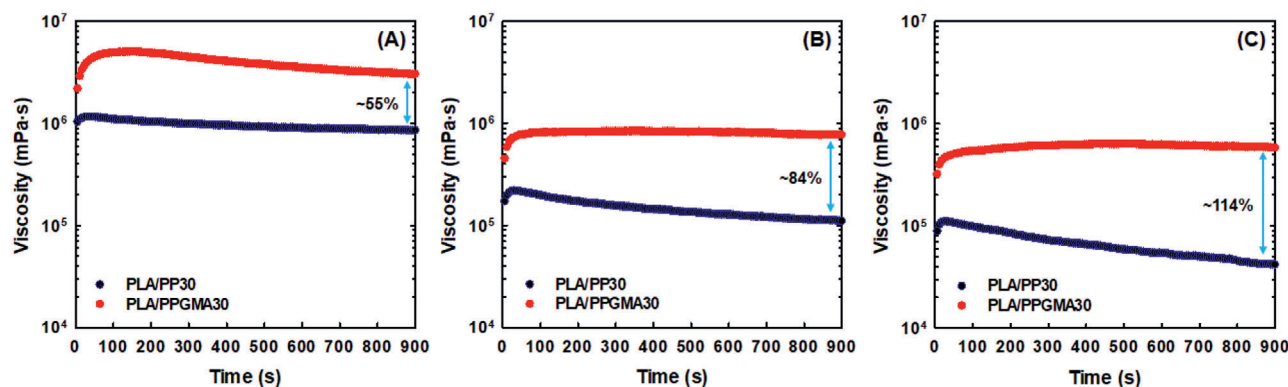


Figure 5. Time-dependent melt viscosity changes of PLA/PPGMA30 and PLA/PP30 blends at different temperatures of A) 180 °C, B) 220 °C, and C) 240 °C.

On the other hand, in the case of the reactive PLA/PPGMA30 blend, the reaction between molecular chains proceeds with time in the molten state, and the higher the temperature, the more accelerated it. As a result, it can be interpreted that the copolymer produced by the reaction between the PLA and PPGMA injection chain has a large molecular weight and increases the compatibility, so that the viscosity increases with time. This result confirms that there exists the reactive compatibilization in PLA/PPGMA blends by forming PPGMA-g-PLA copolymers with higher molecular weights and viscosities, which occurs dominantly at higher temperatures, unlike the immiscible PLA/PP blend.

The small amplitude oscillatory shear tests were carried out for characterizing the linear viscoelastic properties of PLA, PP, PPGMA, and their blends. As a result, the shear storage modulus (G'), shear loss modulus (G''), loss tangent ($\tan \delta$), and complex viscosity (η^*) of PLA, PP, PPGMA, PLA/PP30 blend, and PLA/PPGMA30 blend were measured in the angular frequency range of 0.1–100 rad s^{-1} at 240 °C and shear strain of 2%, as shown in Figure 6. The frequency-dependent shear storage moduli, shear loss modulus, and complex viscosity of the neat PLA were far lower than those of the neat PP and PPGMA, as can be seen in Figure 6A–C, which implies that the chain mobility of PLA was higher than that of PP and PPGMA.^[39] It is interesting that the shear moduli and complex viscosity of the PLA/PPGMA30 blend were found to be much higher than those of the PLA/PP30 blend, although the moduli and viscosity of PPGMA were even lower than those of the neat PP because of the bulky GMA groups grafted on PP chains. This result demonstrates that the PLA/PPGMA30 blend has highly improved compatibility, compared to the PLA/PP30 blend, owing to the specific interactions between PLA and PPGMA as well as the presence of PPGMA-g-PLA copolymers formed during the melt-compounding.^[40]

In the loss tangent ($\tan \delta$) versus frequency curves of Figure 6D, the $\tan \delta$ peak of the PLA/PPGMA30 blend was weaker and appeared at higher frequency, compared to those of the neat PLA and PLA/PP30 blend. It means that the chain mobility of PLA in the PLA/PPGMA30 blend at a melt state was restricted by the presence of PLA segments grafted on PPGMA chains as well as the specific interactions between PLA and PPGMA.

3.4. Thermal Properties

To characterize the effect of PPGMA on the thermal transition behavior of PLA/PPGMA blends, DSC heating and cooling thermograms of PLA, PPGMA, and PLA/PPGMA blends were obtained, as shown in Figure 7, and characteristic transition temperatures are summarized in Table 2. In the DSC heating thermogram (Figure 7A), the glass transition (T_g), cold-crystallization (T_{cc}), and melting (T_{m2}) of the neat PLA were observed at ≈ 56.3 , ≈ 123.5 , and ≈ 148.7 °C, respectively. On the other hand, the melting temperatures (T_{m3}) of the neat PP and PPGMA were detected at ≈ 164.1 and ≈ 161.9 °C, respectively, without noticeable glass transition and cold-crystallization. For the PLA/PPGMA blends, the glass transition (T_g), cold-crystallization (T_{cc}), and three melting temperatures (T_{m1} , T_{m2} , and T_{m3}) were observed at 56.3–58.0, 106.5–108.9, and 144.0–162.5 °C, respectively. The T_{cc} is associated with the cold-crystallization of PLA component in the blends, while the T_{m1} and T_{m2} values are related to multiple melting temperatures of PLA crystals owing to the melting-recrystallization-remelting behavior and T_{m3} value is caused by the melting of PP crystals. It is noticeable that the T_{cc} values of the PLA/PPGMA blends are lower than that of the neat PLA, which is owing to the nucleating effect of the PP crystals on the cold-crystallization of PLA in the blends. Although the overall thermal transition behavior of the PLA/PP30 blend was similar with that of the PLA/PPGMA30 blend, the melting transition (T_{m3}) of PP crystals in the PLA/PPGMA30 was lower and broader than that of the PLA/PP30. This result stems from the melting of relatively thinner PP crystals in the PLA/PPGMA30 blend with enhanced compatibility due to the presence of PPGMA-g-PLA copolymers and/or specific intermolecular interactions of PPGMA and PLA, in comparison to the PLA/PP30 blend. In the DSC cooling thermograms (Figure 7B), the T_{mc} value of the neat PP is ≈ 114.8 °C, while the T_{mc} values of the PLA/PPGMA blends increase by ≈ 19 °C. It is conjectured that the PPGMA-g-PLA copolymers facilitate the heterogeneous nucleation of PP chains, resulting in accelerating the overall melt-crystallization of PP.^[41,42]

The effect of PPGMA content on the thermal decomposition behavior of the PLA/PPGMA blends was investigated by using TGA and DTG thermograms of Figure 8A,B. The neat PLA, PP, and PPGMA show single-step thermal decomposition behavior, while the PLA/PPGMA blends with 1–30 wt% PPGMA

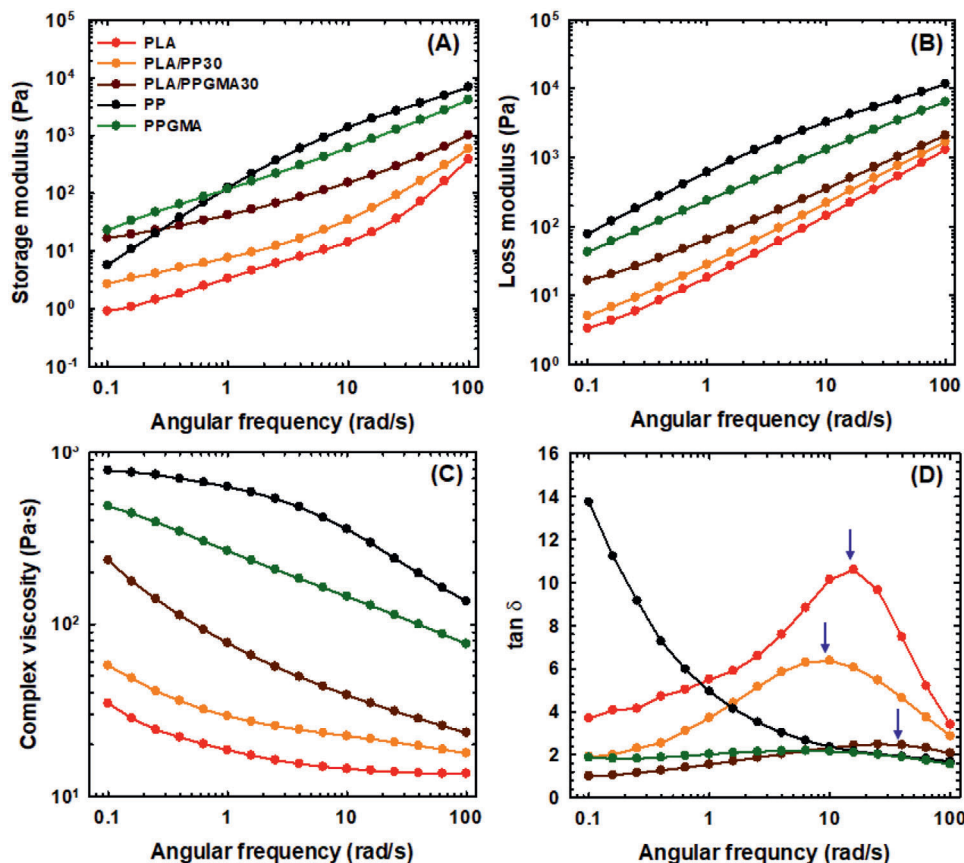


Figure 6. A) Storage modulus, B) loss modulus, C) complex viscosity, and D) $\tan \delta$ of neat PLA, PPGMA, PP, PLA/PPGMA30, and PLA/PP30 as a function of angular frequency.

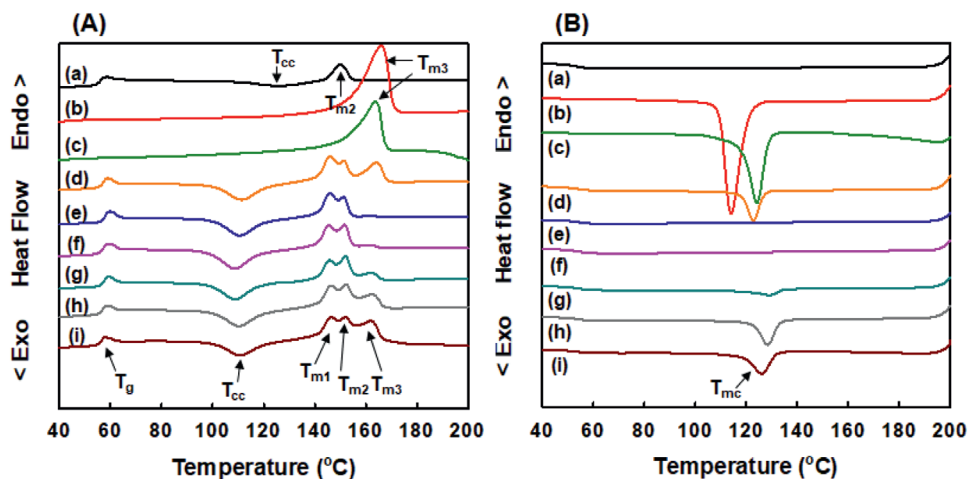


Figure 7. DSC A) heating and B) cooling thermograms of a) PLA, b) PP, c) PPGMA, d) PLA/PP30, e) PLA/PPGMA1, f) PLA/PPGMA5, g) PLA/PPGMA10, h) PLA/PPGMA20, and i) PLA/PPGMA30.

contents display two-step thermal decomposition behavior owing to two components. The thermal decomposition peak temperatures ($T_{d,PLA}$ and $T_{d,PP}$) of PLA and PPGMA components in the blends, which are obtained from the DTG thermograms, are listed in **Table 3**. The $T_{d,PLA}$ value of the neat PLA is measured to

be ≈ 374.6 °C and the $T_{d,PP}$ value of the neat PPGMA is ≈ 428.3 °C, which is ≈ 17.4 °C lower than that of the neat PP. The $T_{d,PLA}$ and $T_{d,PP}$ values of the PLA/PPGMA blends are found to be in the range of 370.2–378.6 and 426.0.2–456.4 °C, respectively, which are higher for the blends with higher PPGMA contents due to

Table 2. Thermal transition temperatures of PLA, PPGMA, PP, and their blends.

Sample code	Heating					Cooling
	$T_{g,PLA}$ [°C]	$T_{cc,PLA}$ [°C]	$T_{m1,PLA}$ [°C]	$T_{m2,PLA}$ [°C]	$T_{m3,PP}$ [°C]	$T_{mc,PP}$ [°C]
PLA	56.3	123.5	—	148.7	—	—
PLA/PPGMA1	58.0	108.9	144.4	149.3	162.5	122.8
PLA/PPGMA5	57.3	107.0	144.1	150.0	160.6	122.7
PLA/PPGMA10	57.5	106.5	144.0	150.4	161.5	133.0
PLA/PPGMA20	57.3	108.2	144.8	150.7	161.8	129.5
PLA/PPGMA30	56.3	108.5	144.4	150.3	161.0	127.8
PLA/PP30	57.0	109.2	144.2	149.9	163.6	123.7
PPGMA	—	—	—	—	161.9	125.0
PP	—	—	—	—	164.1	114.8

the existence of the PPGMA-g-PLA copolymers and the specific interactions between PLA and PPGMA. Consistently, the $T_{d,PP}$ values of PLA/PPGMA20 and PLA/PPGMA30 blends are found to be even higher than those of the neat PP and PLA/PP30 blend. It is thus valid to conclude that the thermal stability of both PLA and PPGMA components in the blends could be improved by the synergistic effects of chemical reaction and specific interactions between PLA and PPGMA at a molecular level.

3.5. Mechanical Properties

To investigate the influences of the PPGMA content on the dynamic mechanical thermal properties of PLA/PPGMA blends, the temperature-dependent storage moduli (E') of PLA, PP, PPGMA, and their blends were obtained, as shown in Figure 9. For the neat PLA, the storage moduli of $\approx 10^9$ Pa at the glassy state decrease abruptly to $\approx 10^6$ Pa at ≈ 70 °C owing to the glass transition and increase to a level of $\approx 10^8$ Pa above 120 °C due to the cold-crystallization. In case of the neat PP and PPGMA, the storage moduli decrease slightly with increasing the temperature because of their semicrystalline features. The overall temperature-dependent changes of storage moduli of the PLA/PPGMA blends with 1–30 wt% PPGMA were similar with the neat PLA. However,

Table 3. Thermal decomposition peak temperatures of PLA, PPGMA, PP, and their blends.

Sample code	$T_{d,PLA}$ [°C]	$T_{d,PP}$ [°C]
PLA	374.6	—
PLA/PPGMA1	370.2	426.0
PLA/PPGMA5	371.4	444.4
PLA/PPGMA10	372.1	440.2
PLA/PPGMA20	378.9	456.4
PLA/PPGMA30	378.3	453.1
PLA/PP30	377.4	428.9
PPGMA	—	428.3
PP	—	445.7

the storage moduli at the rubbery temperature region of 70–100 °C were higher for the PLA/PPGMA blends with higher PPGMA loadings due to the flexibility of PPGMA. In addition, the cold-crystallization-induced storage moduli increment temperatures (≈ 100 °C) of PLA component in the PLA/PPGMA blends were even lower than that (≈ 120 °C) of the neat PLA, which stems from the effective nucleating effects of PP crystals on the cold-crystallization of PLA component in the blends.

For quantitative analysis, the storage moduli of PLA, PPGMA, and their blends at 30 °C were compared, as represented in Figure 9B. It is found that the storage modulus of the neat PLA is far higher than those of the neat PP and PPGMA with flexible and ductile characteristic. In cases of the PLA/PPGMA blends, the storage modulus increases with increasing the PPGMA content up to 20 wt%, but it decreases for the blend with 30 wt% PPGMA content. It is also found that the storage moduli of PLA/PPGMA blends are much higher than that of PLA/PP30 blend. The noticeable increment in the storage moduli for the PLA/PPGMA blends is believed to originate from the enhanced compatibilization effect of PPGMA-g-PLA copolymers and the interfacial adhesion by the specific interactions between PLA and PPGMA.

Figure 9C displays the loss tangent ($\tan \delta$) versus temperature curves for PLA, PP, PPGMA, and their blends with various compositions. The $\tan \delta$ peak temperatures in Figure 9C are all associated with the glass transition, as listed in Table 4. The

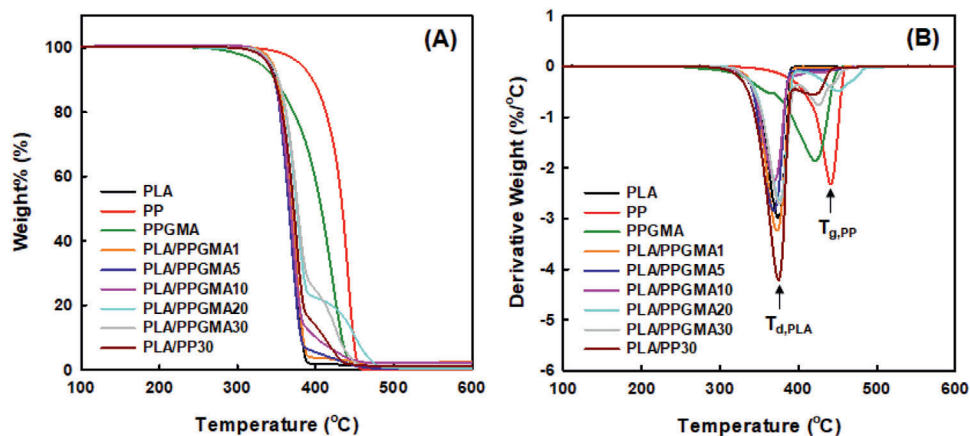


Figure 8. A) TGA and B) DTG thermograms of PLA, PPGMA, PP, and their blends.

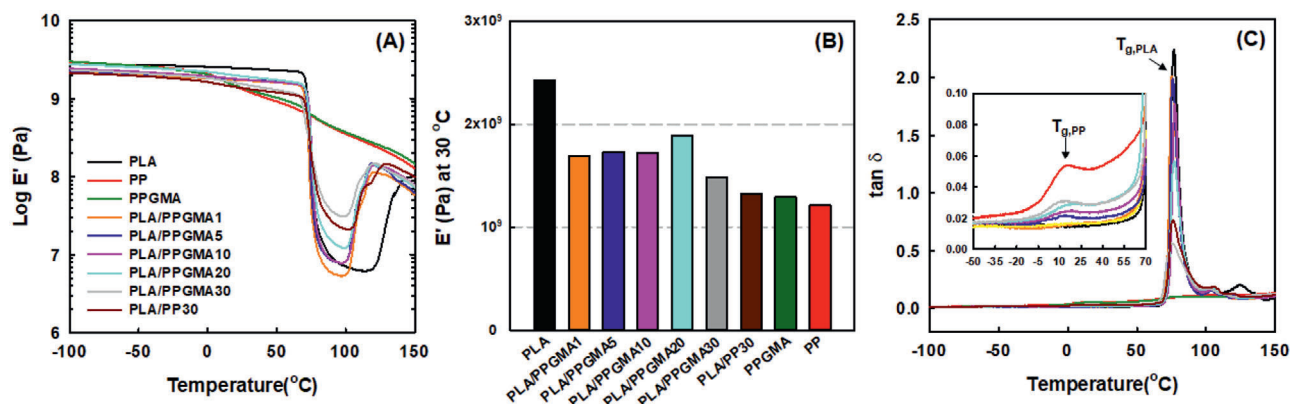


Figure 9. A) Temperature-dependent storage modulus changes, B) storage modulus at 30 °C, and C) temperature-dependent loss tangent changes of PLA, PPGMA, PP, and their blends.

Table 4. Loss tangent ($\tan \delta$) peak temperatures associated with glass transition of PLA, PPGMA, PP, and their blends.

Sample code	$T_{g,PP}$ [°C]	$T_{g,PLA}$ [°C]
PLA	–	76.5
PLA/PPGMA1	10.2	74.8
PLA/PPGMA5	11.4	75.6
PLA/PPGMA10	12.1	77.1
PLA/PPGMA20	14.3	76.4
PLA/PPGMA30	15.1	74.9
PLA/PP 30	9.2	75.1
PPGMA	11.0	–
PP	14.4	–

strong $\tan \delta$ peak temperature ($T_{g,PLA}$) of the neat PLA was ≈ 76.5 °C, whereas the weak and broad $\tan \delta$ peak temperatures ($T_{g,PP}$) of the neat PP and PPGMA were characterized to be ≈ 14.4 and ≈ 11.0 °C, respectively. The relative weak and broad $\tan \delta$ peaks for the neat PP and PPGMA are associated with the semicrystalline features, unlike the amorphous PLA. In addition, the slightly low $\tan \delta$ peak temperature of PPGMA, in compared with PP, is related to the enhanced chain mobility owing to the grafted GMA segments. In cases of the PLA/PPGMA blends, two $\tan \delta$ peak temperatures were detected, as summarized in Table 4. The $\tan \delta$ peak temperatures ($T_{g,PLA}$) of the major PLA component in the blends were almost constant, regardless of the PPGMA content, while the $\tan \delta$ peak ($T_{g,PP}$) of the minor PPGMA component shifted to higher temperatures and its intensity became weaker with the increment of the PPGMA content in the blends. It denotes that the chain mobility of PPGMA is more severely restricted in the blends by the specific interactions between PPGMA and PLA components and the compatibilization of PPGMA-g-PLA copolymers. It is conjectured that the interfacial boundary of the blends with the compatibilizing PPGMA-g-PLA copolymers is strengthened, and the chain mobility of the PPGMA components is reduced due to the efficient reaction and specific interactions with PLA chains in the blends.

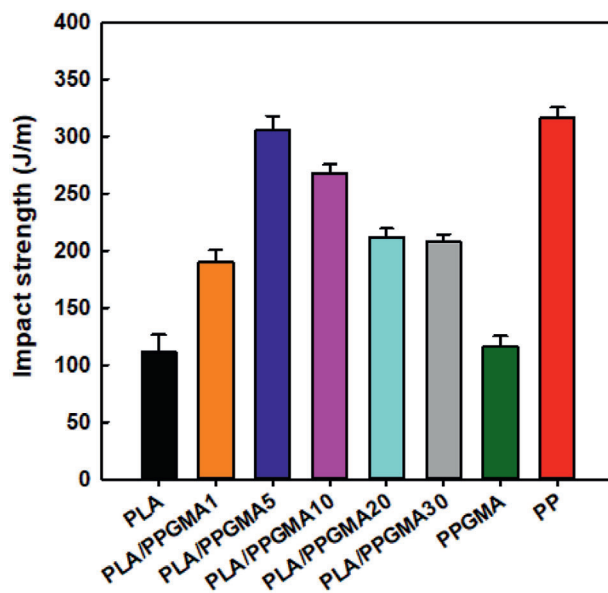


Figure 10. Izod impact strength of PLA, PPGMA, PP, and PLA/PPGMA blends.

For the practical applications of PLA/PPGMA blends, the impact strength tests were carried out and the results are shown in Figure 10. The impact strength of the neat PLA was evaluated to be 111.6 ± 15.0 J m⁻¹, indicating typically brittle fracture behavior. The neat PP was characterized to have high impact strength of 316.9 ± 8.5 J m⁻¹ owing to its ductile mechanical property. On the other hand, the impact strength of the PPGMA was measured to be 116.1 ± 9.5 J m⁻¹, which is caused by the presence of grafted components of GMA and styrene. However, PPGMA is reactive or interactive with PLA. Therefore, in cases of the PLA/PPGMA blends, the impact strength increased from 190.3 ± 10.5 J m⁻¹ for PLA/PPGMA1 to 305.6 ± 12.2 J m⁻¹ for PLA/PPGMA5, but it decreased for the blends with higher PPGMA contents, i.e., 268.0 ± 7.7 J m⁻¹ for PLA/PPGMA10, 212.0 ± 7.5 J m⁻¹ for PLA/PPGMA20, and 208.1 ± 6.7 J m⁻¹ for PLA/PPGMA30. Due to the trade-off effect of the high reactivity and low toughness of PPGMA component, the highest impact strength was attained



for the PLA blend with 5 wt% PPGMA. For comparison, the impact strength of PLA/PP5 was tested and measured to be $113.1 \pm 11.2 \text{ J m}^{-1}$, which was slightly higher than that ($\approx 111.6 \text{ J m}^{-1}$) of the neat PLA, but far lower than that ($\approx 305.6 \text{ J m}^{-1}$) of PLA/PPGMA5. The low impact strength of PLA/PP5 blend is related to the incompatibility and weak interfacial adhesion between PP domains and PLA matrix, unlike the PLA/PPGMA5 blend. Overall, it is believed that the PLA/PPGMA5 blend with a maximum impact strength of $\approx 305.6 \text{ J m}^{-1}$, which is almost three times higher than the neat PLA and is comparable to the neat PP, can be utilized as highly toughened PLA-based material for potential commercial applications requiring high environmental, thermal, and mechanical performance.^[43]

4. Conclusions

In this study, the morphological features, rheological, thermal, and mechanical properties of PLA/PPGMA blends, which were fabricated by facile and mass processible reactive melt-compounding, were investigated by considering the PPGMA content, reactivity, and intermolecular interactions. The SEM images confirmed that the PLA/PPGMA30 blend has a relatively uniform and ductile fractured morphology without showing phase-separated domains and interfaces, unlike the PLA/PP30 blend. The FT-IR spectra revealed the presence of specific interactions between the epoxy/ester groups of PPGMA and the hydroxyl/carboxyl/ester groups of PLA as well as the possibility of the formation of PPGMA-g-PLA copolymers in the PLA/PPGMA blends during the melt-compounding. It was supported by the rheological analyses that the melt viscosity and shear moduli of PLA/PPGMA30 blend were far higher than those of PLA/PP30 blend, although the shear moduli of PPGMA were even lower than those of the neat PP. The DSC heating and cooling thermograms of the PLA/PPGMA blends showed that the melt-crystallization of PPGMA was retarded, while the cold-crystallization of PLA was accelerated by the nucleating effect of PP crystals. The thermal decomposition peak temperatures of PLA and PPGMA components in the blends were measured to be 370.2–378.6 and 426.0–456.4 °C, respectively, which were higher for the blends with higher PPGMA contents. Furthermore, the thermal decomposition temperatures of the PPGMA component in the PLA/PPGMA20 and PLA/PPGMA30 blends were even higher than those of the neat PP and PLA/PP30 blend. The DMA results showed that the storage modulus of the PLA/PPGMA blends at 30 °C increased with increasing the PPGMA content up to 20 wt%, but it decreased for the blend with 30 wt% PPGMA content. Nonetheless, the storage moduli of the PLA/PPGMA blends at 30 °C were even higher than those of neat PP, PPGMA, and PLA/PP30 blend. In addition, the impact strength ($\approx 305.6 \text{ J m}^{-1}$) of the PLA blend with 5 wt% PPGMA was measured to be almost three times higher than the neat PLA ($\approx 111.6 \text{ J m}^{-1}$) and comparable to the neat PP ($\approx 316.9 \text{ J m}^{-1}$). Overall, it is reasonable to contend that highly tough, thermally stable, and process-efficient PLA/PPGMA blends without significant stiffness loss can be utilized as ecofriendly and sustainable polymeric materials for a variety of commodity and engineering applications.

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Conflict of Interest

The authors declare no conflict of interest.

Author Contributions

The manuscript was written through the contributions of all the authors. All authors have given approval to the final version of the manuscript.

Data Availability Statement

Data sharing is not applicable to this article as no new data were created or analyzed in this study.

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